Results and Discussion (Working Draft)

Deep Reinforcement Learning for Optimal Order Execution on Real Hyperliquid Order-Book Data

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Working draft: every result produced so far is included at this stage; the selection of material and the prose will be revised in later dissertation stages as the remaining pre-registered experiments complete.

1 Experimental architecture

1.1 Two complementary tracks

The study asks whether deep reinforcement learning (DRL) agents can execute a large buy order in Bitcoin more cheaply than time-weighted average price (TWAP) scheduling, and it answers this on two complementary experimental tracks built from the same underlying market: Hyperliquid BTC-USD order-book data from December 2025.

1. **The frozen-replay track.** Agents trade against replayed historical order-book snapshots (1-minute and 10-second bars). The market contains every empirical predictive signal that was present in the data, but it cannot react to the agent: the agent's own trades do not move the replayed prices. This track therefore measures whether *signal-timing* skill exists, under the assumption of negligible market impact.
2. **The reactive-simulator track.** Agents trade inside a queue-reactive model (QRM) calibrated on per-order (L4) BTC data, in which the agent's market orders consume real book liquidity, move the price, and interact with calibrated refill dynamics. Two regime-specific calibrations are used (calm and volatile). This track measures whether *impact-management* skill exists, in a market that reacts.

Together the tracks cover the two mechanisms by which an execution agent could beat TWAP; each track holds the other's mechanism fixed. The reactive-simulator track is the study's main focus; the frozen-replay track is a secondary strand whose chief role is to motivate the move to a reacting market (Section 6). A pre-registered extension that combines both mechanisms in one environment (the measured-signal extension, Section 8) is designed and queued.

1.2 Environment validation (reactive simulator)

Before any agent was trained, the reactive simulator had to pass pre-registered validation gates. Table 1 reports the gates of record after the background-drift correction described below: the agent's own impact is real and persistent on the relevant horizon (an identical order submitted immediately after the first costs roughly 2.4× as much, because the first has consumed the visible liquidity and the book has not yet refilled), dump costs are size-monotone, the book demonstrably refills, simulated cost-versus-size growth tracks the real books within the pre-registered band, and the TWAP benchmark completes 100% of episodes.

A material confound was found and corrected at this stage: the December 2025 calibration month was directionally rising, and replaying it naïvely gave every fast-buying policy a mechanical advantage. The background move process was therefore drift-neutralised (volatility preserved), and a permanent fairness gate was added: the residual drift is statistically zero in both regimes, and no constant-pace policy is significantly cheaper than TWAP (maximum $|t| = 1.77$ across all pace multiples and regimes). Figure 1 shows the before-and-after effect on an identical 20-run design retrained under the corrected process: the apparent calm-regime advantage shrinks from -0.019 bps to an immaterial -0.006 bps, statistically indistinguishable from zero across seeds (95% confidence interval $[-0.020, +0.010]$) and an order of magnitude below the 0.05 bps economic materiality floor adopted below.

Table 1: Environment validation gates of record after drift correction. Rows one to three verify that the agent's own trading genuinely moves the simulated market and that the book both remembers and recovers from the impact; row four that simulated cost growth with order size matches the real order book; rows five and six that the TWAP benchmark operates correctly and that immediate execution is properly penalised; the final two rows that the corrected background process gives no schedule a mechanical advantage. Archived records: `step4_gates_v3.json` and `step3g/fairness_verdict_*.json`, in which these checks carry the internal identifiers G1′, G2, G3 and the fairness gate, in row order.

regime	check	measured	pass band	verdict
calm	own impact is real and persists (2nd immediate dump / 1st, cost ratio)	2.41	≥ 1.25	PASS
calm	dump cost increases with order size (Spearman ρ)	1.00	> 0	PASS
calm	book refills after impact (probe cost +30s vs +1s, bps)	-0.46 vs 0.35	lower at +30s	PASS
calm	cost-vs-size growth matches the real book (sim vs real ratio)	2.45 vs 2.46	within band	PASS
calm	fixed-TWAP completes every episode	100%	$\geq$ 99%	PASS
calm	dumping costs more than scheduling (dump / TWAP, drift-free, bps)	0.40 / 0.20	dump $\geq$ TWAP	PASS
calm	no residual background drift (ticks/episode; t)	0.12 (t=0.27)	t n.s.	PASS
calm	no constant-pace policy beats TWAP (max $\lvert t\rvert$ across paces)	1.42	none significant	PASS
volatile	own impact is real and persists (2nd immediate dump / 1st, cost ratio)	2.35	≥ 1.25	PASS
volatile	dump cost increases with order size (Spearman ρ)	1.00	> 0	PASS
volatile	book refills after impact (probe cost +30s vs +1s, bps)	-0.23 vs 0.14	lower at +30s	PASS
volatile	cost-vs-size growth matches the real book (sim vs real ratio)	2.44 vs 1.99	within band	PASS
volatile	fixed-TWAP completes every episode	100%	$\geq$ 99%	PASS
volatile	dumping costs more than scheduling (dump / TWAP, drift-free, bps)	0.24 / 0.05	dump $\geq$ TWAP	PASS
volatile	no residual background drift (ticks/episode; t)	0.07 (t=0.06)	t n.s.	PASS
volatile	no constant-pace policy beats TWAP (max $\lvert t\rvert$ across paces)	1.77	none significant	PASS

1.3 Evaluation machinery

Paired evaluation with common random numbers (CRN). Every policy (agent and both benchmarks) is evaluated on the same 2,000 episodes with identical episode seeds, so market luck cancels in the paired per-episode differences and the comparison isolates policy skill.

Two TWAP benchmarks, both required. The fixed benchmark is the textbook open-loop schedule (order/N per decision, with fractional-unit carry). The adaptive benchmark re-plans the remaining quantity over the remaining time each second; it is exactly the agent's "always 1.0× pace" action, so it is the neutral element of the agent's own action space and the industry-realistic implementation. The pre-registered edge definition requires beating *both*; the adaptive benchmark

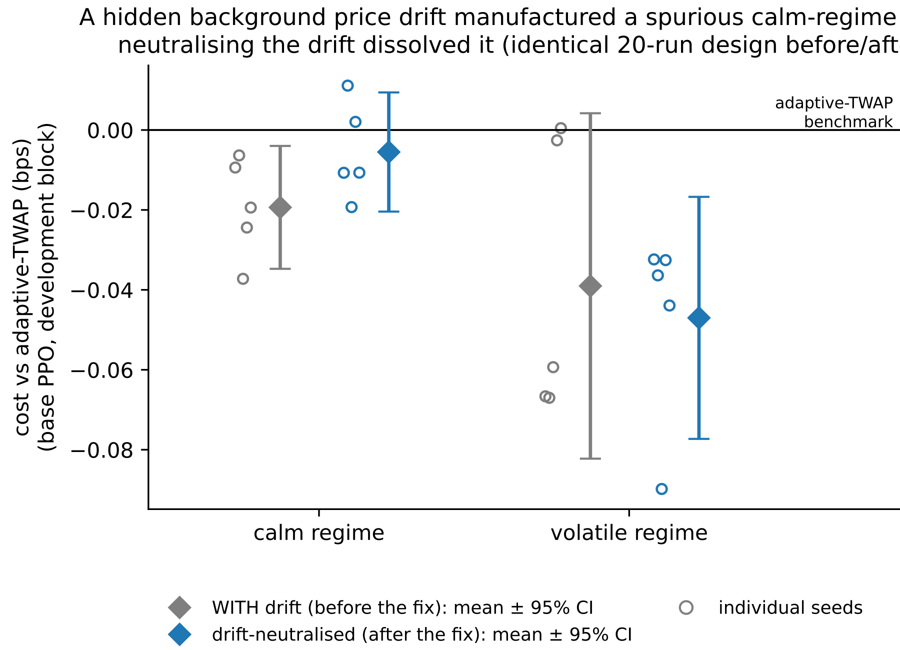

Figure 1: The drift confound. The identical 20-run base campaign evaluated with the original background process (grey) and after drift-neutralisation (blue); the design is identical and the agents are retrained under each background process. Open circles are individual seeds, diamonds the group mean with 95% confidence interval, and the horizontal line marks adaptive-TWAP parity. After neutralisation the apparent calm-regime advantage shrinks from −0.019 bps to an immaterial −0.006 bps, statistically indistinguishable from zero (the 95% confidence interval spans zero) and an order of magnitude below the pre-registered 0.05 bps materiality floor, while the volatile group moves within seed noise. The residual-drift and constant-pace checks of Table 1 confirm the corrected background gives no schedule a mechanical advantage.

is the headline metric because it is the stricter of the two. At the primary setting the choice is empirically immaterial, with the two benchmarks agreeing to within 0.001 basis points (bps) per seed. At the smallest order size in the robustness grid (5 BTC) the two schedules genuinely diverge, by up to 0.08 bps per seed, which is one more reason the edge definition requires beating both. No verdict in this document changes under either benchmark.

Frozen decision rules. The definition of "beats TWAP" was frozen before any run: per-seed Wilcoxon signed-rank $p < 0.01$ against both benchmarks, the sign consistent in at least four of five seeds, and a pooled advantage of at least 0.05 bps (an economic materiality floor, since CRN pairing can make trivially small differences statistically significant; 0.05 bps is approximately 1% of one taker fee). A lower investigation tripwire (pooled ≤ -0.02 bps with across-seed $p < 0.05$) guarantees that a small-but-consistent effect is never silently discarded: it triggers a pre-registered follow-up instead.

Three-layer evaluation. Because roughly 200 trained configurations were screened against the same development episodes, the design inserts a replication layer between screening and confirmation, analogous to discovery and replication cohorts in genomics or public and private leaderboards in machine-learning competitions: (1) *screen* on the development block; (2) *replicate* candidates on a never-before-used reserve block after escalating to five seeds; (3) *confirm* survivors on a sealed block used exactly once. Figure 2 summarises the architecture and what happened at each layer. The replication layer was added, dated and documented, after the first two sealed tests failed; its first use (Section 3) justified it emphatically.

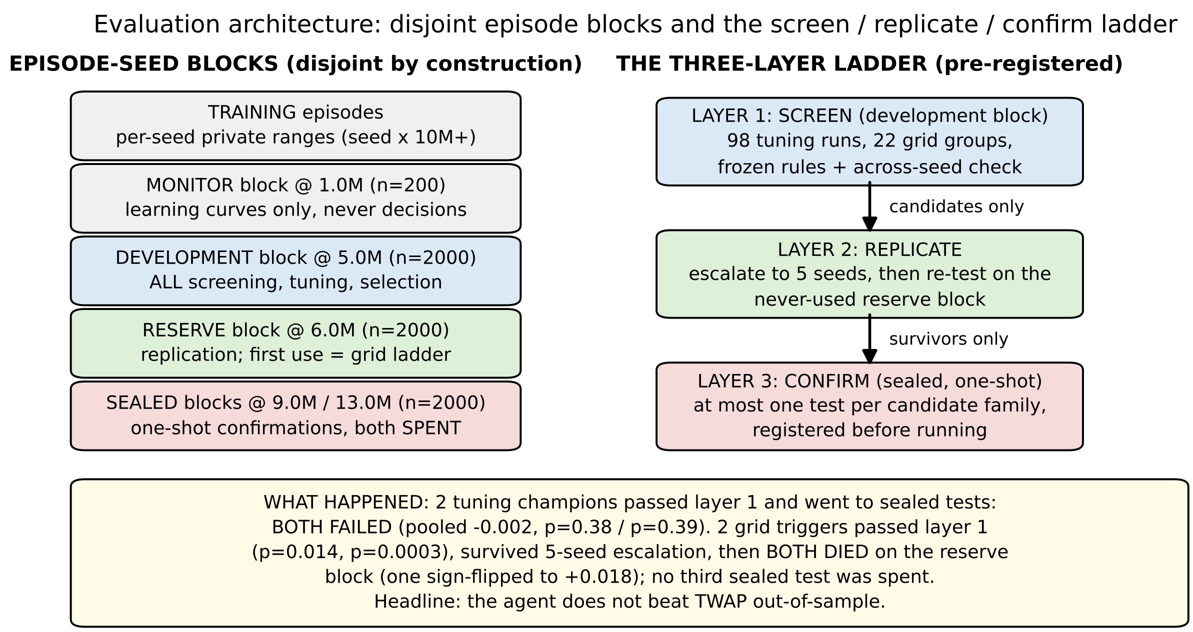

Figure 2: Evaluation architecture. Each evaluation block is a disjoint set of episode random seeds, so no two blocks share market paths. Roughly 200 trained configurations were screened on the development block; groups meeting the pre-registered trigger were escalated to five seeds and replicated on the never-before-used reserve block; only selection winners reached the single-use sealed blocks. The diagram records what happened at each layer.

2 Research question 1: can DRL beat TWAP in a reacting market?

We trained PPO and DQN agents to buy 25 BTC in five minutes inside the reactive simulator, tuned them extensively, and sent the two best configurations to one-shot sealed tests. Neither survived. The headline result of this study is a *boundary null*: a development-block advantage that repeatedly fails to exist on fresh data.

2.1 Primary campaign

Twenty agents (PPO and DQN, calm and volatile regimes, five seeds each; 2M training steps; identical 30×5 networks and matched gradient sample throughput of ∼20M sample-passes per run) were evaluated under the frozen rules. Table 2 gives every run. No cell met the edge definition. All ten PPO agents passed the behaviour audit; seven of ten DQN agents collapsed (Section 5). The volatile-regime PPO group showed the strongest sub-threshold signal: five of five seeds cheaper than both benchmarks, pooled -0.047 bps, across-seed $p = 0.006$. Figure 3 shows the contrast directly. For base PPO and the later-selected variant alike, the calm-regime seeds cluster at the benchmark line while the volatile-regime seeds sit consistently below it; Section 7 returns to this regime split once the confirmation verdicts are in hand.

Table 2: Primary campaign, all 20 runs. Source: `step5_v3/`.

algo	regime	seed	vs fixed-TWAP (bps)	p_{fix}	vs adaptive-TWAP (bps)	valid
dqn	calm	0	+0.1124	0.015	+0.1125	**NO**
dqn	calm	1	+0.0596	0.072	+0.0597	**NO**
dqn	calm	2	+0.0263	0.326	+0.0264	**NO**
dqn	calm	3	+0.1533	0.000	+0.1534	**NO**
dqn	calm	4	-0.0016	0.829	-0.0015	yes
dqn	volatile	0	+0.4176	0.001	+0.4169	**NO**
dqn	volatile	1	-0.0296	0.632	-0.0303	yes
dqn	volatile	2	+0.3017	0.003	+0.3011	**NO**
dqn	volatile	3	-0.0389	0.383	-0.0396	yes
dqn	volatile	4	-0.0197	0.893	-0.0203	**NO**
ppo	calm	0	-0.0108	0.958	-0.0107	yes
ppo	calm	1	+0.0110	0.576	+0.0111	yes
ppo	calm	2	-0.0194	0.324	-0.0193	yes
ppo	calm	3	+0.0020	0.597	+0.0020	yes
ppo	calm	4	-0.0108	0.632	-0.0107	yes
ppo	volatile	0	-0.0317	0.350	-0.0324	yes
ppo	volatile	1	-0.0357	0.442	-0.0364	yes
ppo	volatile	2	-0.0892	0.012	-0.0898	yes
ppo	volatile	3	-0.0319	0.417	-0.0326	yes
ppo	volatile	4	-0.0433	0.444	-0.0439	yes

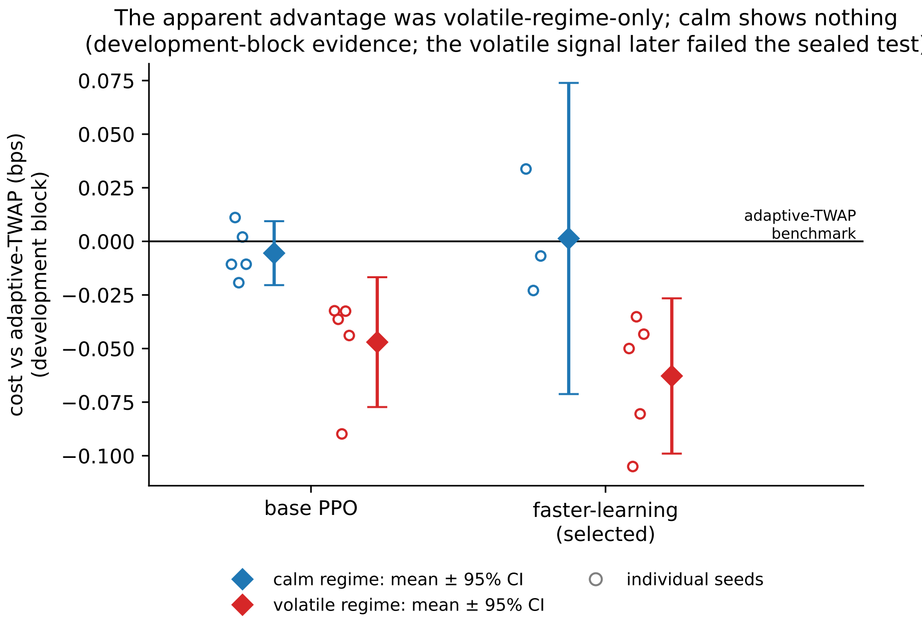

Figure 3: Regime dependence of the development-block signal (research question 2). Base PPO and the selected variant evaluated per regime on the development block; open circles are individual seeds, diamonds the group mean with 95% confidence interval. The apparent advantage was volatile-regime-only and later failed both sealed tests; the calm regime shows nothing after the drift fix.

2.2 Hyperparameter search and selection

Twelve pre-registered PPO variants and two DQN collapse-remediation variants were screened (98 runs in total; Table 3 summarises the groups; the full run-level record is Appendix Table A.1; configurations in Table 10). In the volatile regime seven of eight healthy variants reproduced a small consistent advantage (Figure 4); in the calm regime nothing was consistent (Figure 5). The selected configuration (learning rate 10^{-3}, all else base) pooled -0.063 bps with across-seed $p = 0.0043$ on five seeds.

2.3 Sealed confirmations: the headline

Two configurations earned sealed tests. The first (learning rate 10^{-3}) was chosen on volatile-regime performance, the metric used during screening. A documented audit of the selection step then found that the pre-registered selection rule, applied exactly as written (ranking configurations on performance across both regimes jointly), would have chosen a different configuration (network 128×128); that configuration received the second sealed test, so both readings of the selection rule were confirmed out of sample. Each champion was retrained on five fresh seeds and evaluated once on its own never-used sealed block. **Both failed** (Table 4, Figure 6): pooled -0.0023 bps ($p = 0.379$) and -0.0022 bps ($p = 0.394$), confidence intervals bracketing zero. The development-block signal, consistent across five seeds and twelve variants, does not exist out of sample. The two-test family was closed by pre-registration; the selection-metric question is resolved empirically, since both candidate champions returned approximately zero.

Undertraining does not explain the failures. Figure 7 tracks both configurations across the full 2M-step training budget on the independent monitor block: both oscillate slightly above TWAP

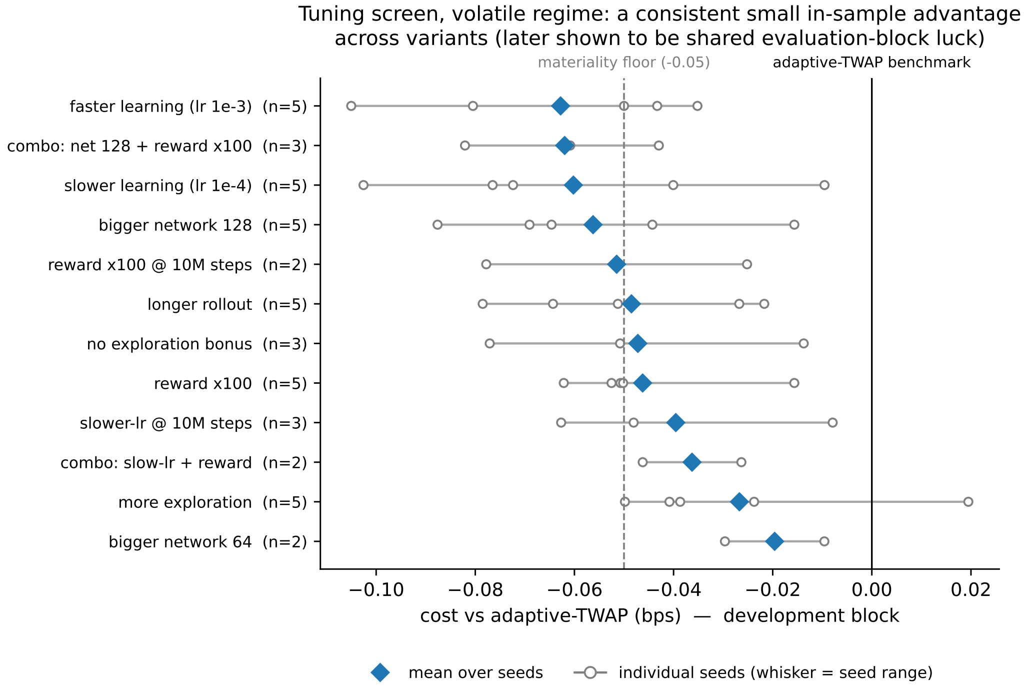

Figure 4: Tuning screen, volatile regime. Each row is one pre-registered variant, labelled by its single change from the base configuration; open circles are individual seed means on the development block, the whisker spans the seed range, the diamond is the group mean, and the dashed line is the -0.05 bps materiality floor. Seven of eight healthy variants sit below zero, an in-sample consistency later shown to be shared evaluation-block luck.

parity within seed noise, with no sustained improvement trend in the second half of training. The policies had converged, and the monitor block never showed the advantage that the development block did; in hindsight this was the first sign that the advantage belonged to the evaluation episodes rather than to the policies.

2.4 Per-episode cost distributions

The sealed records store per-run means and significance values; to characterise the full cost distributions, the twenty primary-campaign agents were replayed deterministically on the same 2,000 CRN episodes, storing every episode's implementation shortfall for the agent and both benchmarks. The replay's integrity is verifiable: the recomputed paired means equal the sealed judgement records exactly for all twenty runs.

Two facts stand out (Figure 8, Figure 9, Table 5). First, the agents' absolute cost distributions are visually indistinguishable from the benchmarks': execution cost is dominated by market conditions, not by policy, and the volatile regime widens every distribution by roughly a factor of 2.5 (standard deviation 2.35 to 5.88 bps, Table 5). Second, the paired per-episode differences dwarf their means: the strongest development-block signal (-0.047 bps in the volatile regime) is three orders of magnitude smaller than the episode-level spread of the paired differences, which is precisely why 2,000 paired episodes and strict multiple-testing discipline are necessary in this domain.

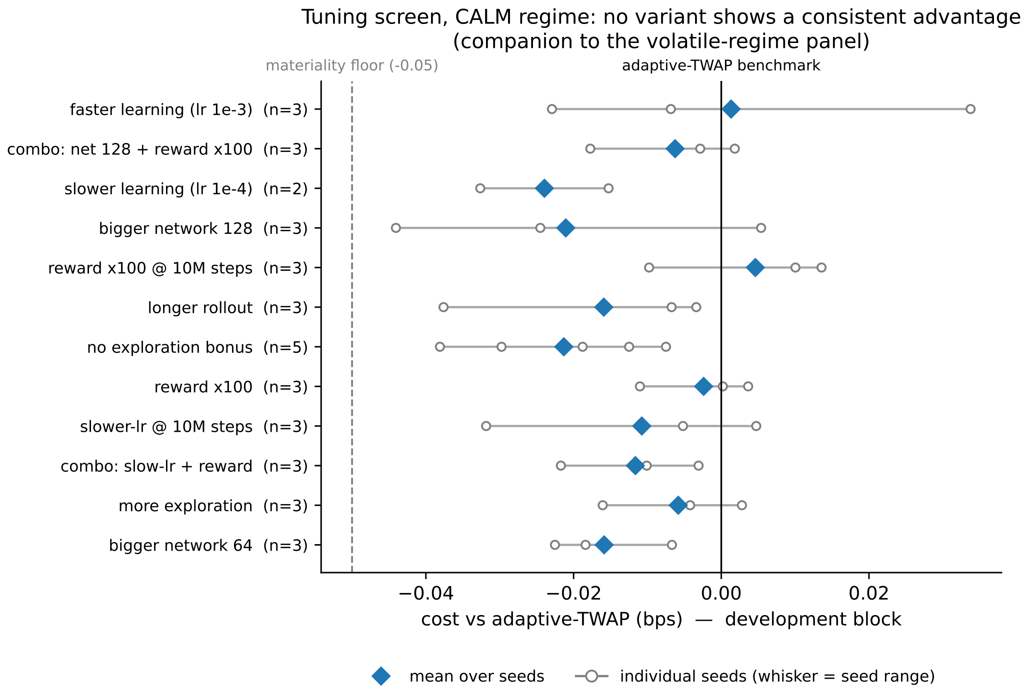

Figure 5: Tuning screen, calm regime, same layout as Figure 4. No variant shows a consistent advantage; seed ranges straddle the benchmark throughout.

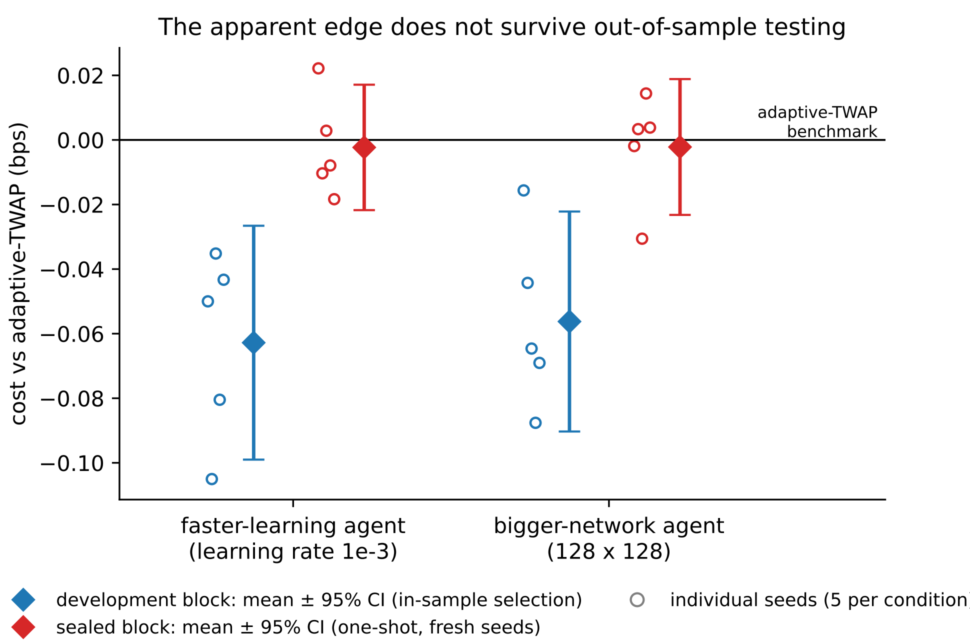

Figure 6: The headline result. For each champion configuration, the development-block evaluation that selected it (blue) is shown next to its one-shot sealed evaluation on fresh seeds and a never-used episode block (red); open circles are individual seeds, diamonds the mean with 95% confidence interval, the horizontal line adaptive-TWAP parity. Both sealed results sit at approximately zero, so the development-block advantage does not exist out of sample.

Table 3: Tuning and selection, all 28 groups. Source: `step5_selection_v3/`.

variant	change	regime	valid/total seeds	pooled vs adaptive (bps)	across-seed p
_v3a	learning rate 1e-3	volatile	5/5	-0.0628	0.004
_v3b	learning rate 1e-4	volatile	5/5	-0.0602	0.010
_v2	reward x100	volatile	5/5	-0.0462	0.002
_v1b	network 128x128	volatile	5/5	-0.0562	0.005
_v5	n_steps 8192	volatile	5/5	-0.0485	0.006
_v4a	ent_coef 0.0	volatile	3/3	-0.0472	0.062
_v4b	ent_coef 0.05	volatile	5/5	-0.0267	0.048
_v1a	network 64x64	volatile	2/3	-0.0196	0.151
_v6	reward x100 @ 10M	volatile	2/3	-0.0515	0.150
_v6b	lr 1e-4 @ 10M	volatile	3/3	-0.0395	0.069
_w3a	net128 + reward x100	volatile	3/3	-0.0620	0.016
_w3b	lr 1e-4 + reward x100	volatile	2/3	-0.0363	0.085
_d1	DQN eps 0.05+anneal	volatile	1/3	-0.0403	–
_d2	DQN reward x100+net64	volatile	3/3	-0.0259	0.199
_v3a	learning rate 1e-3	calm	3/3	+0.0013	0.528
_v3b	learning rate 1e-4	calm	2/3	-0.0239	0.111
_v2	reward x100	calm	3/3	-0.0024	0.322
_v1b	network 128x128	calm	3/3	-0.0211	0.140
_v5	n_steps 8192	calm	3/3	-0.0159	0.141
_v4a	ent_coef 0.0	calm	5/5	-0.0213	0.010
_v4b	ent_coef 0.05	calm	3/3	-0.0058	0.200
_v1a	network 64x64	calm	3/3	-0.0159	0.040
_v6	reward x100 @ 10M	calm	3/3	+0.0046	0.705
_v6b	lr 1e-4 @ 10M	calm	3/3	-0.0108	0.214
_w3a	net128 + reward x100	calm	3/3	-0.0063	0.200
_w3b	lr 1e-4 + reward x100	calm	3/3	-0.0116	0.083
_d1	DQN eps 0.05+anneal	calm	1/3	-0.0068	–
_d2	DQN reward x100+net64	calm	0/3	–	–

Table 4: Both sealed confirmations, per seed, with the pass/fail checklist. Source: `step5_confirm_v3a/`, `step5_confirm_v1b/`.

config	sealed block	seed	vs fixed	vs adaptive	p (adaptive)	valid
faster-learning (v3a)	9,000,000	5	+0.0221	+0.0222	0.749	yes
		6	-0.0104	-0.0103	0.813	yes
		7	+0.0028	+0.0028	0.804	yes
		8	-0.0080	-0.0079	0.936	yes
		9	-0.0184	-0.0184	0.758	yes
	pooled vs adaptive = -0.0023 bps; across-seed p = 0.379; cheaper in 3/5; **PASS = FALSE**					
bigger-network (v1b)	13,000,000	10	-0.0010	-0.0019	0.650	yes
		11	+0.0042	+0.0033	0.953	yes
		12	-0.0297	-0.0306	0.294	yes
		13	+0.0153	+0.0144	0.491	yes
		14	+0.0047	+0.0038	0.961	yes
	pooled vs adaptive = -0.0022 bps; across-seed p = 0.394; cheaper in 2/5; **PASS = FALSE**					

Figure 7: Training curves in the volatile regime, evaluated on the independent monitor block (200 fixed episodes) throughout the 2M-step budget; lines are means over five seeds, bands one standard deviation across seeds. Both configurations hover slightly above TWAP parity with no sustained improvement trend, so the sealed failures are not explained by undertraining.

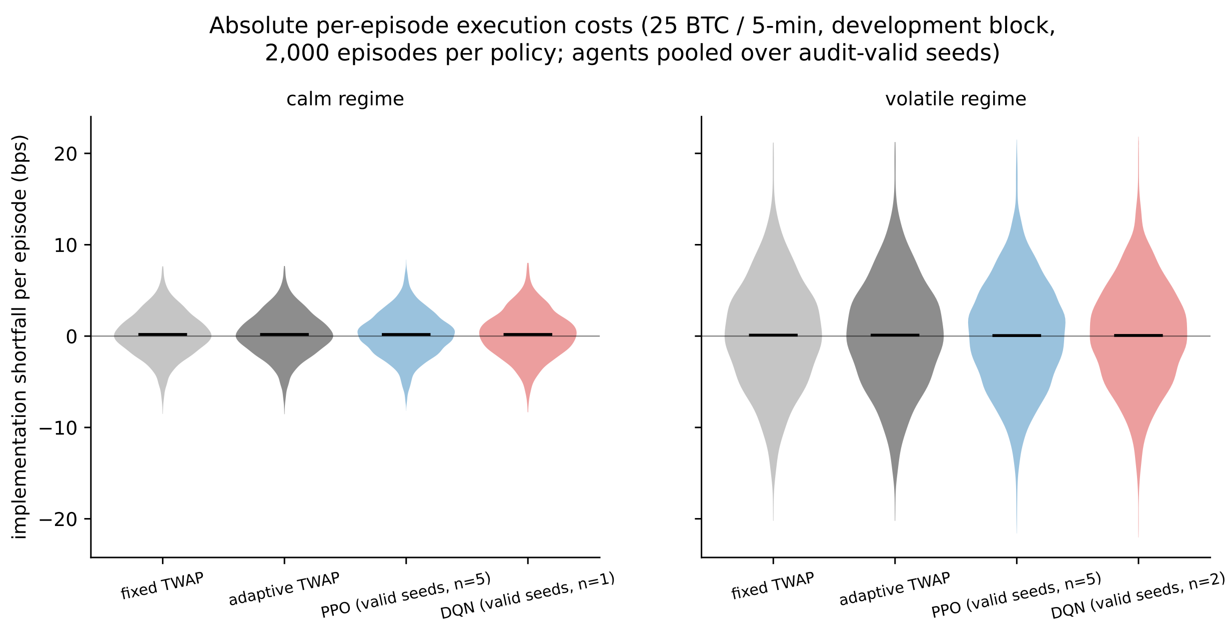

Figure 8: Absolute per-episode execution costs at the primary setting (development block; 2,000 episodes per policy; deterministic replay whose recomputed means match the sealed records exactly). Each violin is one policy's full cost distribution. Agents and benchmarks are visually indistinguishable, and the volatile regime widens every distribution by roughly a factor of 2.5; market conditions, not policy choice, dominate execution cost.

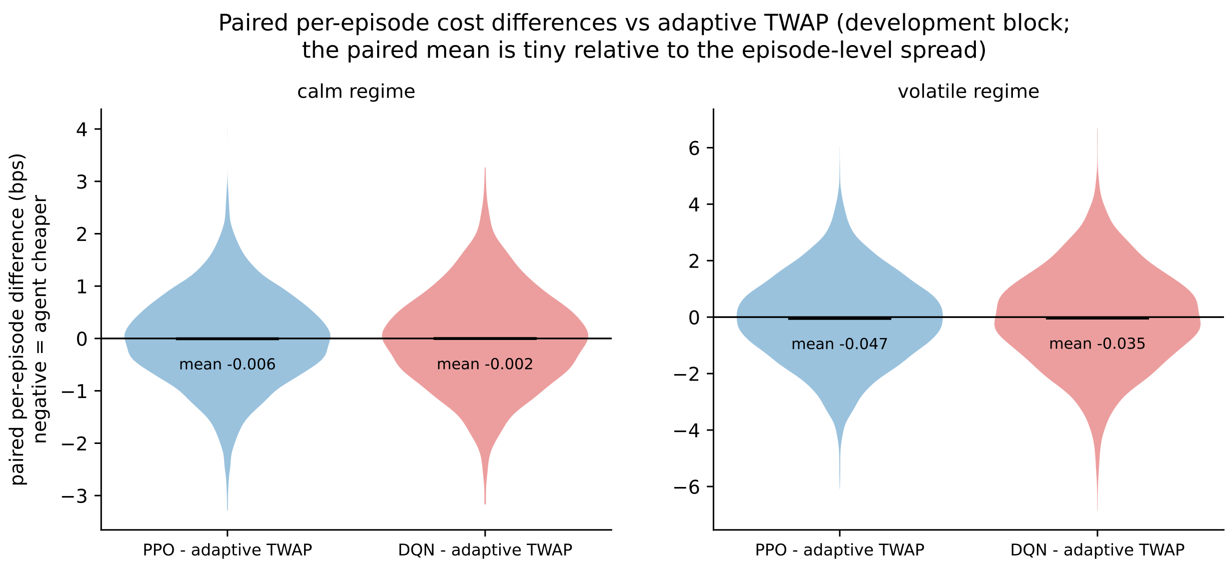

Figure 9: Paired per-episode cost differences against adaptive TWAP on identical episodes (negative = agent cheaper). The episode-level spread is roughly three orders of magnitude larger than the mean differences under test, which is why 2,000 paired episodes and strict significance discipline are needed to measure anything in this domain.

Table 5: Descriptive statistics of per-episode costs at the primary setting (bps; reactive track, development block). Source: `per_episode_v3/`.

regime	policy	mean	sd	median	5%	95%	n episodes
calm	fixed TWAP	+0.1803	2.3546	+0.1840	-3.7238	+4.0427	2,000
calm	adaptive TWAP	+0.1802	2.3540	+0.1810	-3.7212	+4.0245	2,000
calm	PPO (valid seeds)	+0.1747	2.3661	+0.2089	-3.7477	+3.9982	10,000
calm	DQN (valid seeds)	+0.1787	2.4817	+0.1849	-3.9363	+4.2143	2,000
volatile	fixed TWAP	+0.1031	5.8800	+0.1013	-9.7930	+9.4489	2,000
volatile	adaptive TWAP	+0.1037	5.8793	+0.0922	-9.7133	+9.4541	2,000
volatile	PPO (valid seeds)	+0.0567	5.7882	+0.1640	-9.3214	+9.2684	10,000
volatile	DQN (valid seeds)	+0.0688	5.6446	+0.1161	-9.0810	+9.1627	4,000

3 Robustness of the null across the design space

A null at one order size and one deadline could be a blind spot, so the null was stress-tested across a 4×4 grid of order sizes (5 to 50 BTC) and deadlines (2.5 to 20 minutes), in both regimes. It held everywhere; the two cells that looked like exceptions died on fresh data, exactly as the earlier illusion had.

3.1 Order-size response and the full grid

A genuine impact-management advantage should grow with order size. It does not: the development-block advantage is non-monotone in size (absent at 5, 12.5 and 50 BTC, present only at the 25 BTC development size; Figure 10), which is the signature of selection on noise rather than of skill. The full grid (66 new runs; Figure 11, Table 6) is null in 20 of 22 newly tested groups, with several groups significantly *worse* than TWAP (for example 12.5 BTC at the 20-minute deadline in the volatile regime, pooled +0.164 bps).

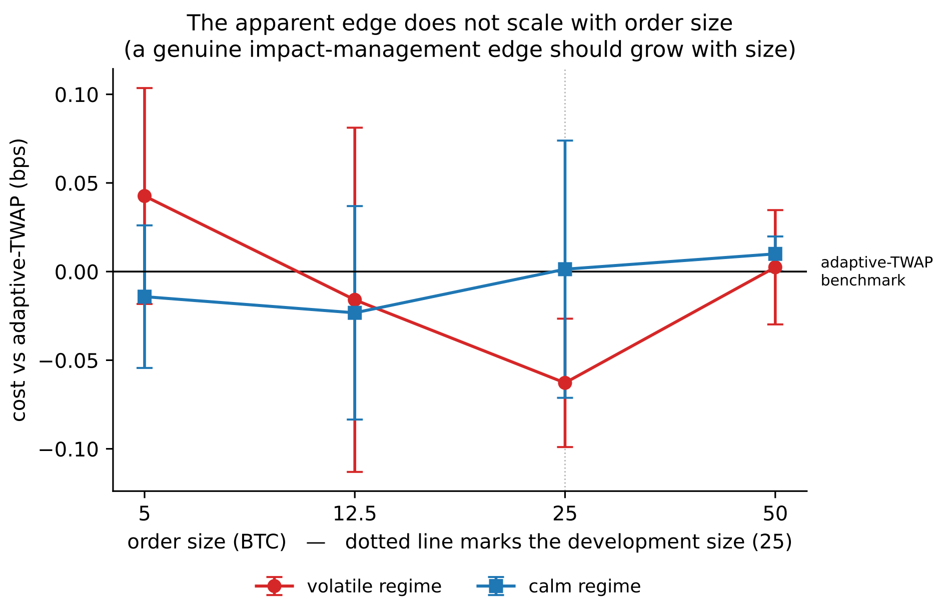

Figure 10: Size response of the development-block signal at the 5-minute deadline; points are group means with 95% confidence intervals per order size and regime, and the dotted line marks the 25 BTC development size. A genuine impact-management advantage should grow with order size; instead the advantage appears only at the size the configuration was developed at, the signature of selection on noise.

3.2 The two triggered cells and the replication ladder

Two calm-regime groups met the pre-registered follow-up trigger: 50 BTC at the 10-minute deadline (pooled −0.054, $p = 0.014$) and 25 BTC at the 20-minute deadline (pooled −0.063, $p = 0.0003$, all three seeds within 0.005 bps of one another). Both survived escalation to five seeds on the development block ($p = 0.0037$ and $p = 0.0001$). Both then **failed the first-ever use of the reserve block**: −0.0095 ($p = 0.17$) and +0.0177 ($p = 0.96$), the second reversing sign entirely (Figure 12, Table 7). No sealed test was spent; the null verdict is final across the design space.

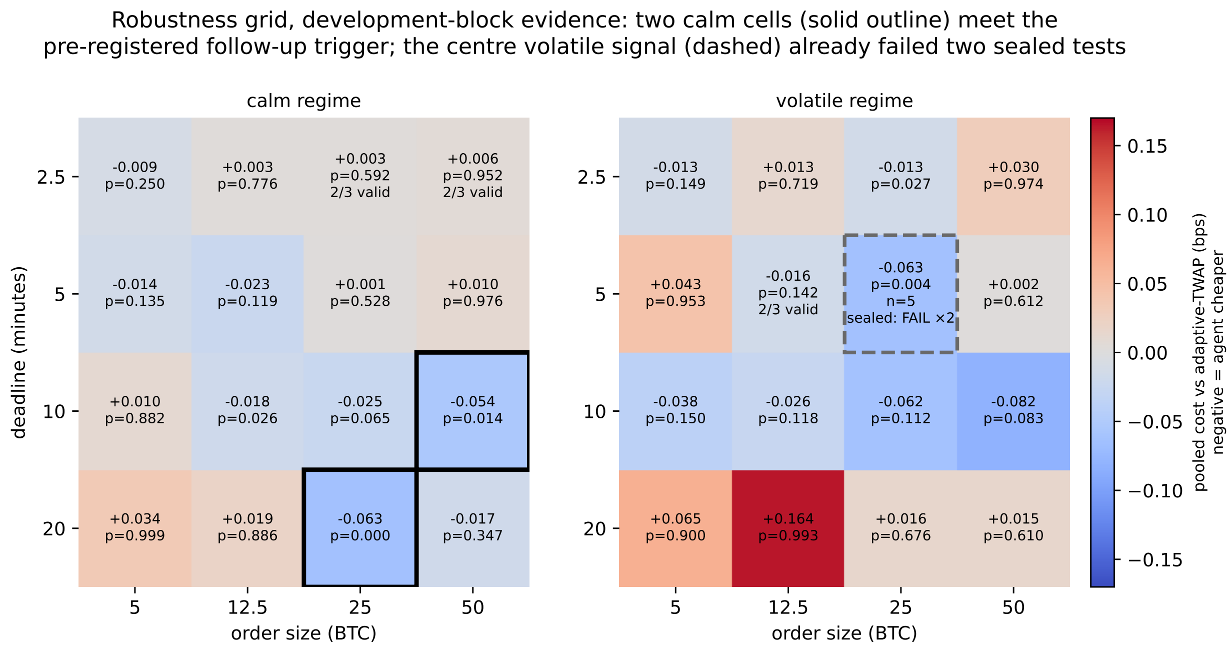

Figure 11: The complete size-by-deadline grid on the development block. Each cell reports the pooled agent cost versus adaptive TWAP (negative = agent cheaper) with its across-seed p-value. Solid outlines mark the two calm cells that met the pre-registered follow-up trigger, both of which subsequently died on the reserve block (Figure 12); the dashed outline marks the volatile centre cell whose signal had already failed both sealed tests.

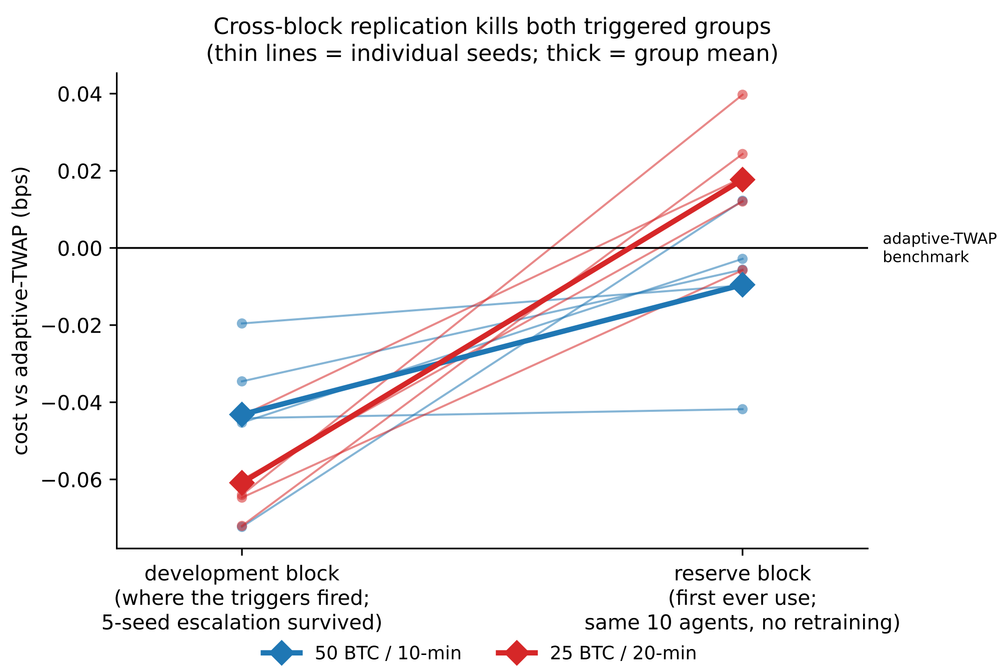

Figure 12: Cross-block replication kills both triggered groups. The same ten escalated agents are evaluated on the development block, where the triggers fired and the five-seed escalation survived, and then on the first-ever use of the reserve block, with no retraining; thin lines are individual seeds, thick lines group means. One group shrinks to insignificance and the other reverses sign entirely.

Table 6: Robustness grid, all 16 cells per regime. Source: `step5_grid_*`, `step5_sweep_*`, `step5_selection_v3`.

regime	deadline (min)	size (BTC)	valid/ total	cheaper	pooled vs adaptive (bps)	across-seed p	follow-up trigger
calm	2.5	5	3/3	2/3	-0.0094	0.2498	no
	2.5	12.5	3/3	1/3	+0.0032	0.7761	no
	2.5	25	2/3	1/2	+0.0029	0.5919	no
	2.5	50	2/3	0/2	+0.0060	0.9525	no
	5	5	3/3	2/3	-0.0142	0.1346	no
	5	12.5	3/3	2/3	-0.0233	0.1191	no
	5	25	3/3	2/3	+0.0013	0.5279	no
	5	50	3/3	0/3	+0.0100	0.9757	no
	10	5	3/3	1/3	+0.0105	0.8819	no
	10	12.5	3/3	3/3	-0.0178	0.0261	no
	10	25	3/3	3/3	-0.0250	0.0646	no
	10	50	3/3	3/3	-0.0539	0.0140	**YES**
	20	5	3/3	0/3	+0.0338	0.9991	no
	20	12.5	3/3	1/3	+0.0189	0.8863	no
	20	25	3/3	3/3	-0.0629	0.0003	**YES**
	20	50	3/3	2/3	-0.0169	0.3466	no
volatile	2.5	5	3/3	2/3	-0.0132	0.1492	no
	2.5	12.5	3/3	1/3	+0.0126	0.7186	no
	2.5	25	3/3	3/3	-0.0134	0.0273	no
	2.5	50	3/3	0/3	+0.0301	0.9740	no
	5	5	3/3	0/3	+0.0426	0.9526	no
	5	12.5	2/3	2/2	-0.0159	0.1424	no
	5	25	5/5	5/5	-0.0628	0.0043	closed (failed both sealed tests)
	5	50	3/3	1/3	+0.0024	0.6120	no
	10	5	3/3	2/3	-0.0383	0.1503	no
	10	12.5	3/3	3/3	-0.0262	0.1180	no
	10	25	3/3	2/3	-0.0619	0.1123	no
	10	50	3/3	3/3	-0.0821	0.0833	no
	20	5	3/3	0/3	+0.0648	0.8996	no
	20	12.5	3/3	0/3	+0.1638	0.9926	no
	20	25	3/3	2/3	+0.0156	0.6764	no
	20	50	3/3	1/3	+0.0150	0.6099	no

Table 7: The replication ladder, per seed. Source: `step5_esc_*`, `step5_xblock_*`.

group	seed	development (bps)	reserve (bps)	change
50 BTC / 10-min	0	-0.0453	-0.0028	+0.0425
	1	-0.0723	+0.0122	+0.0846
	2	-0.0441	-0.0418	+0.0023
	3	-0.0346	-0.0056	+0.0290
	4	-0.0196	-0.0097	+0.0099
	pooled	-0.0432	-0.0095	+0.0336
25 BTC / 20-min	0	-0.0601	+0.0120	+0.0721
	1	-0.0640	+0.0397	+0.1037
	2	-0.0647	-0.0058	+0.0589
	3	-0.0721	+0.0244	+0.0964
	4	-0.0435	+0.0182	+0.0617
	pooled	-0.0609	+0.0177	+0.0786

4 Anatomy of a spurious edge

Three times in this study, results that would pass conventional reporting standards (consistent across seeds, statistically significant, economically material) turned out to be artifacts of the specific evaluation data. Documenting how they arose, and how the pre-registered machinery caught each one, is a central contribution of this dissertation.

1. **The drift confound** (Figure 1): an environment-construction artifact that rewarded fast buying. Caught by an adversarial pre-launch audit; fixed; guarded by a permanent fairness gate.

2. **The volatile development-block edge**: five seeds, twelve corroborating variants, $p = 0.006$. Dead on two independent sealed blocks; also contradicted by the independent monitor block (Figure 13), where the same configurations read *worse* than TWAP. Seed agreement cannot detect shared evaluation-set luck, because every seed is graded on the same episodes.

3. **The calm grid triggers**: the most extreme form, $p = 0.0001$ with a $[-0.069, -0.057]$ confidence interval, reversing to $+0.018$ on the next block (Figure 12). The replication layer caught for the cost of four training runs what would otherwise have consumed a third sealed test.

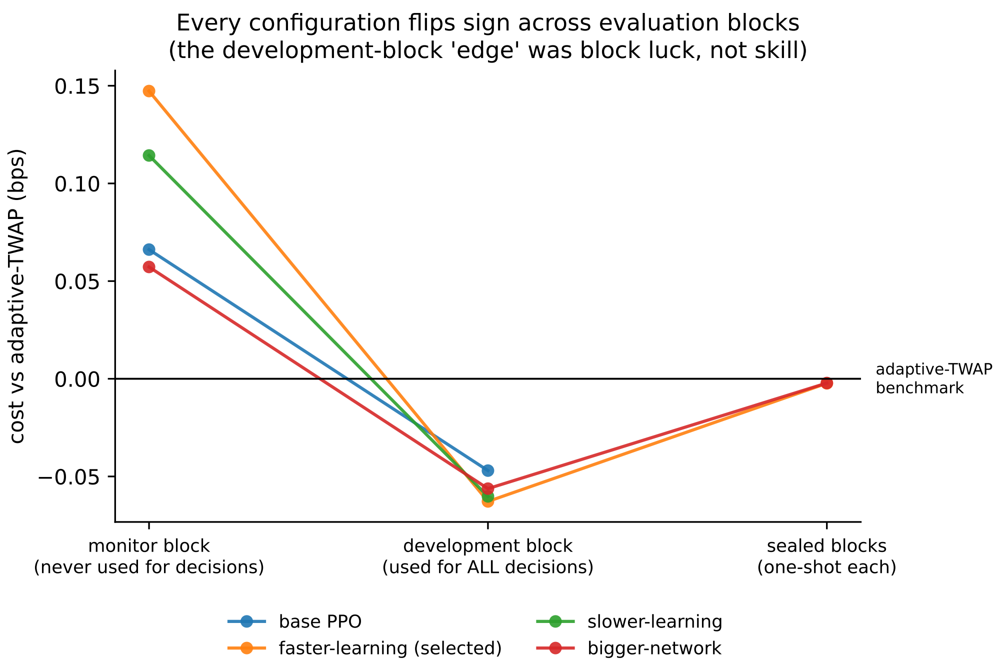

Figure 13: The same four configurations evaluated on disjoint episode blocks flip sign, indicating evaluation-block luck rather than skill. Two lines stop at the development block because sealed blocks are single-use and the pre-registered confirmation budget allowed exactly two tests. Both tests went to the configurations that won selection (faster-learning and bigger-network), so the base and slower-learning configurations were never evaluated on sealed data.

The methodological statement the study supports: in DRL evaluation with expensive runs and few seeds, *across-seed consistency on a fixed evaluation set is not evidence of generalisation*; only fresh evaluation data is. The practical prescription is the three-layer architecture of Figure 2.

5 Why DQN fails: a systematic, diagnosed collapse

DQN did not merely underperform: at realistic order sizes it frequently stopped trading and let the environment's forced deadline purchase execute the order at the worst possible moment. This failure was mapped across settings, its obvious explanations were ruled out with pre-registered experiments, and its mechanism was located in the learned value functions.

The behaviour audit (applied before any cost comparison) flags an agent whose forced deadline purchase completes more than 10% of episodes or whose policy is more than 90% inaction. At the primary setting seven of ten DQN agents collapsed (Figure 14). A pre-registered probe then retrained DQN at three further settings to map the failure, and the pattern is driven by order size, not by deadline. At 5 BTC most agents remain healthy, with 4 of 6 valid. At 25 BTC every agent collapsed even at the shortest 2.5-minute deadline, 0 of 6 valid, and a majority collapsed at every other 25 BTC deadline (Figure 15, Table 8). The collapse also concentrates in the calm regime, with 1 of 9 calm probe agents valid versus 5 of 9 volatile.

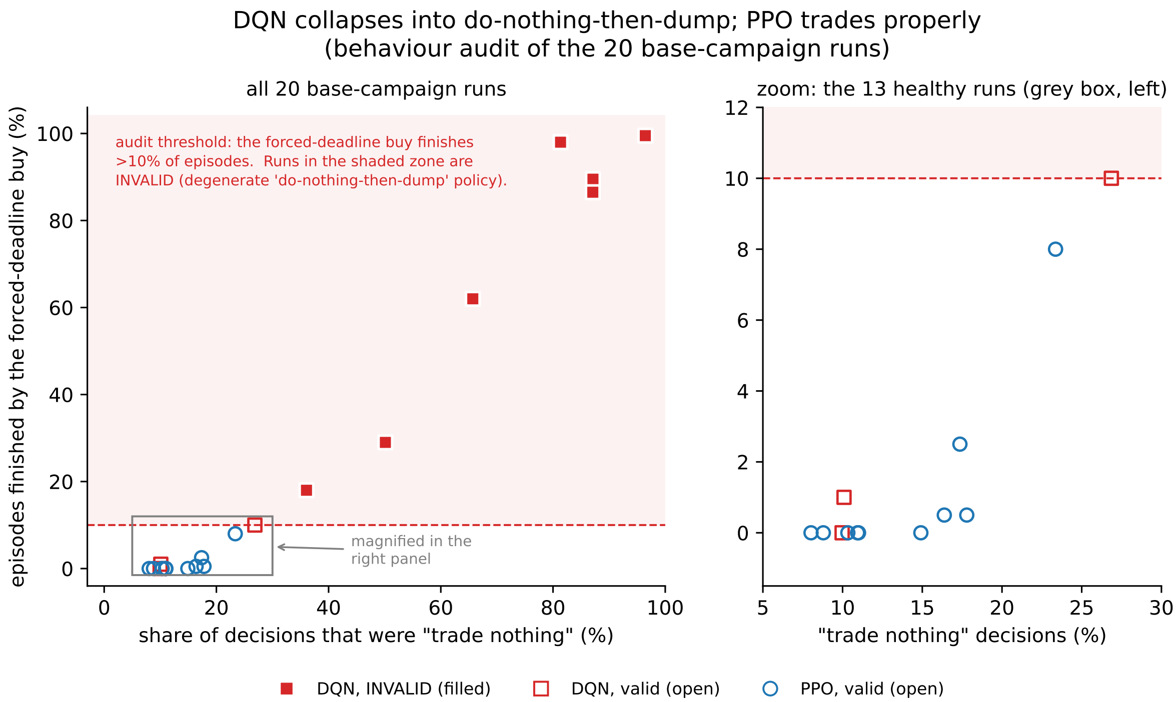

Figure 14: The behaviour audit at the primary setting, all 20 base-campaign runs. Each point is one run. The horizontal axis measures the behaviour itself, the share of individual decisions at which the agent chose to trade nothing; the vertical axis measures its consequence, the share of episodes in which the leftover inventory had to be bought by the forced deadline purchase. The two are related but distinct, since an agent can defer early and still finish on its own. Filled markers breach the 10% audit threshold (shaded region) and are excluded from cost comparisons as invalid. Seven of ten DQN runs collapse into inaction followed by the forced purchase; all ten PPO runs trade properly. The right panel magnifies the healthy cluster.

Two configuration explanations were tested and ruled out. First, two pre-registered fixes (stronger exploration; reward rescaling with a wider network) did not reliably cure the collapse (2/6 and 3/6 valid, calm still 0/3). Second, the update rhythm: our DQN takes approximately 20,000 large gradient updates (batch 1024) where the library default takes 500,000 small ones, with total gradient sample throughput matched ($\sim$20M sample-passes, the same as PPO). Retraining with the library-

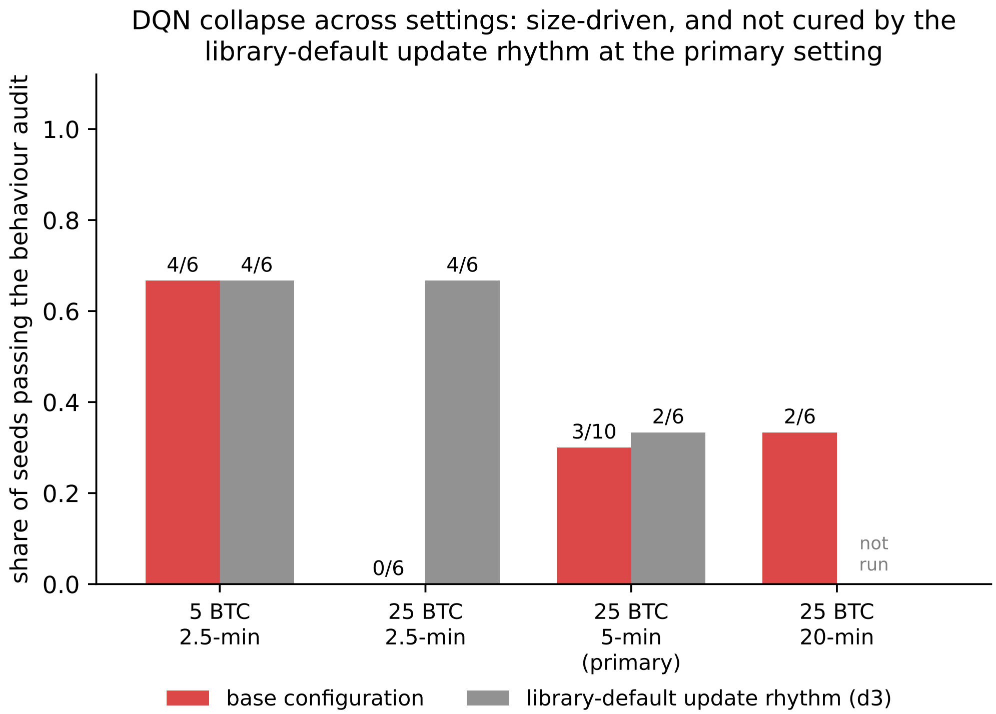

Figure 15: DQN collapse across settings. Bars give the share of seeds passing the behaviour audit at each tested order size and deadline, for the base configuration and for the library-default update rhythm (d3). Collapse worsens with order size at every deadline and is not cured by changing the update rhythm.

default rhythm at three settings (18 runs, pre-registered) left the primary setting unchanged (2/6 valid) while reducing severity only at the most extreme deadline; the collapse is not an update-rhythm artifact (Figure 15, grey bars).

The mechanism was located by reading the trained networks directly (Figure 16). Across *all* probe agents, the learned action-value separations (best minus worst action per state) lie between 0.006 and 0.012, an order of magnitude below the per-step reward noise (standard deviation 0.025 to 0.057 bps). At that scale, which action ranks first is decided by noise and by the reward structure rather than by learned differences. What distinguishes collapsed agents is where the near-flat surface tips: they rank "trade nothing" first in 48% of states on average, against 14% for healthy agents. The interpretation offered (clearly labelled as interpretation): PPO trains on batch-normalised advantages, so microscopic per-step differences still produce full-strength gradients, whereas DQN regresses raw-scale action values, leaving the argmax to noise and to the terminal-penalty structure. The claim is scoped: this is a reproducible failure mode of this value-based configuration on impact-dominated rewards, not a proof that no value-based method can execute (published Double-DQN execution agents exist; testing such upgrades is named future work).

Table 8: DQN cross-setting probe, all 18 runs. Source: `step5_dqnprobe_*`.

cell	regime	seed	do-nothing share	forced-buy share	vs adaptive (bps)	p	audit
5 BTC / 2.5-min	calm	0	93%	43%	-0.0287	0.425	**COLL.**
5 BTC / 2.5-min	calm	1	36%	22%	-0.0250	0.159	**COLL.**
5 BTC / 2.5-min	calm	2	6%	2%	+0.0036	0.824	valid
5 BTC / 2.5-min	volatile	0	7%	0%	+0.0316	0.309	valid
5 BTC / 2.5-min	volatile	1	43%	0%	-0.0540	0.176	valid
5 BTC / 2.5-min	volatile	2	1%	0%	-0.0406	0.041	valid
25 BTC / 2.5-min	calm	0	82%	100%	+0.0258	0.500	**COLL.**
25 BTC / 2.5-min	calm	1	57%	18%	-0.0089	0.646	**COLL.**
25 BTC / 2.5-min	calm	2	13%	67%	-0.0223	0.179	**COLL.**
25 BTC / 2.5-min	volatile	0	17%	48%	-0.0410	0.529	**COLL.**
25 BTC / 2.5-min	volatile	1	46%	12%	-0.0523	0.247	**COLL.**
25 BTC / 2.5-min	volatile	2	37%	72%	-0.0106	0.828	**COLL.**
25 BTC / 20-min	calm	0	43%	54%	-0.0518	0.088	**COLL.**
25 BTC / 20-min	calm	1	43%	87%	-0.0019	0.866	**COLL.**
25 BTC / 20-min	calm	2	54%	70%	-0.0462	0.346	**COLL.**
25 BTC / 20-min	volatile	0	16%	8%	-0.2288	0.025	valid
25 BTC / 20-min	volatile	1	30%	0%	-0.0539	0.427	valid
25 BTC / 20-min	volatile	2	25%	78%	-0.0193	0.570	**COLL.**

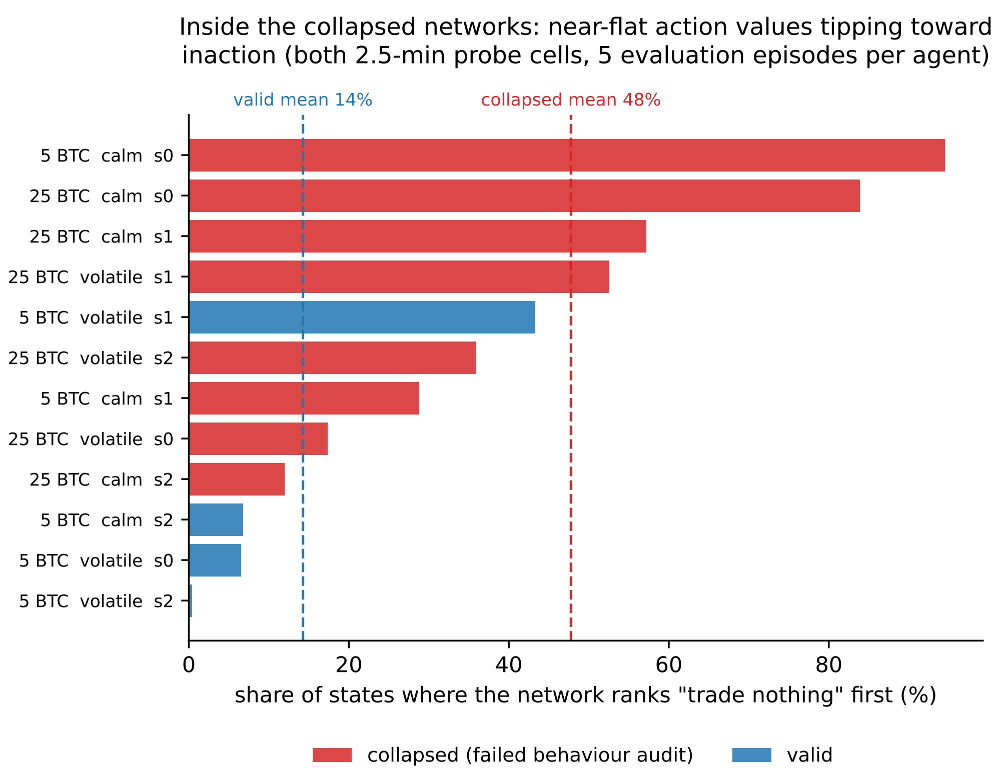

Figure 16: Inside the trained DQN networks. For each probe agent, the share of sampled states in which the learned action values rank "trade nothing" first; collapsed agents average 48%, valid agents 14%. The learned value separations (0.006 to 0.012) sit an order of magnitude below the per-step reward noise, so this near-flat value surface tips easily into systematic inaction.

6 The frozen-replay track: validation results

On replayed historical data, where every real predictive signal is present but the market cannot react, no agent beats TWAP consistently on any of the three design levers (bar resolution, order size, deadline). This is the study's secondary track; the reactive-simulator track of Sections 2 to 5 is the main focus. These are validation-set numbers by design; the track's one-shot sealed exam is built and gated, and it is the track's final verdict.

All 70 agents were evaluated on 400 validation episodes against TWAP on identical episodes. They form 14 arms, an arm being one combination of dataset, algorithm and order size, trained at five seeds each (Table 9, Figure 17). One arm is notably agent-favourable (PPO at 96.57 BTC on 1-minute bars, pooled -0.089 bps, 4/5 seeds cheaper); with 14 arms screened, approximately one such arm is expected by chance, and it receives no special treatment: every arm sits the same sealed exam. On 10-second bars at the 30-minute deadline agents are consistently *worse* than TWAP. The DQN deadline-leaning failure appears on this track too (seven flagged runs; the L2 mirror of Section 5). The size lever (Figure 18) shows benchmark costs growing steeply with order size while no exploitable agent advantage opens at any trained size. Full per-run detail is Appendix Table A.2.

Two verification results from this track are reported. First, a completeness census of all 70 agents, run before the sealed exam, found that four runs of one arm had terminated early at 200,000 of the 2,000,000 training steps without saving final models. They were excluded, the arm's provisional validation figure was withdrawn, and the four seeds were retrained at full budget; the corrected estimate (-0.030 bps) is three times weaker than the provisional one, which is precisely why the census gate exists. Second, the track's training pipeline was verified to be bit-exactly reproducible: an agent retrained from scratch reproduced its saved counterpart exactly, with an identical configuration, an identical validation number to 17 decimal places, and zero difference across the entire training curve. The sealed-exam evaluator of Section 8 is held to the same standard.

Table 9: Frozen-replay track validation summary, all 14 arms. Source: the saved agents' validation re-evaluation.

panel	arm	pooled vs TWAP (bps)	cheaper seeds	flagged seeds	across-seed p
1-min bars, 30-min	PPO 96.57	-0.0890	4/5	0/5	0.037
1-min bars, 30-min	PPO 193.13	-0.0297	3/5	0/5	0.167
1-min bars, 30-min	DQN 96.57	+0.0925	3/5	1/5	0.724
1-min bars, 30-min	DQN 193.13	+0.0463	3/5	2/5	0.738
10-s bars, 30-min	PPO 96.57	+0.1779	0/5	0/5	1.000
10-s bars, 30-min	PPO 193.13	+0.0740	1/5	0/5	0.781
10-s bars, 30-min	PPO 386.27	+0.2010	0/5	0/5	1.000
10-s bars, 30-min	DQN 96.57	+0.7661	0/5	1/5	0.884
10-s bars, 30-min	DQN 193.13	+0.1419	1/5	0/5	0.940
10-s bars, 30-min	DQN 386.27	+0.2043	1/5	1/5	0.981
10-s bars, 10-min	PPO 96.57	+0.0066	3/5	0/5	0.562
10-s bars, 10-min	PPO 193.13	+0.0324	1/5	0/5	0.797
10-s bars, 10-min	DQN 96.57	+0.0247	3/5	1/5	0.674
10-s bars, 10-min	DQN 193.13	+0.1151	2/5	1/5	0.872

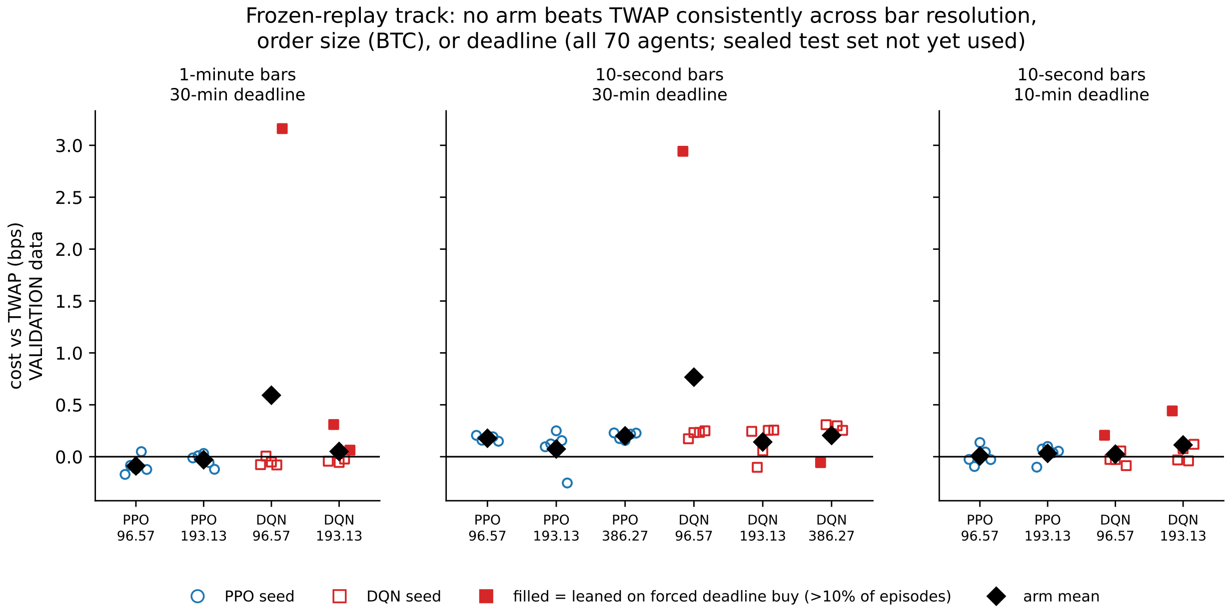

Figure 17: Frozen-replay track, the three-lever null. Each panel varies one design lever (bar resolution, order size, deadline); points are per-seed validation results against TWAP (negative = agent cheaper), and filled markers flag runs that leaned on the forced deadline purchase, the frozen-replay mirror of the DQN failure. No arm beats TWAP consistently on any lever. Validation data; the one-shot sealed exam is the track's final verdict.

Status: A planted-signal credibility check was performed during this track's audit phase: episodes containing a deliberately injected directional edge confirmed that the evaluation correctly rewards policies that exploit it. The output files of that check were not preserved, so the check will be regenerated before its result is cited quantitatively.

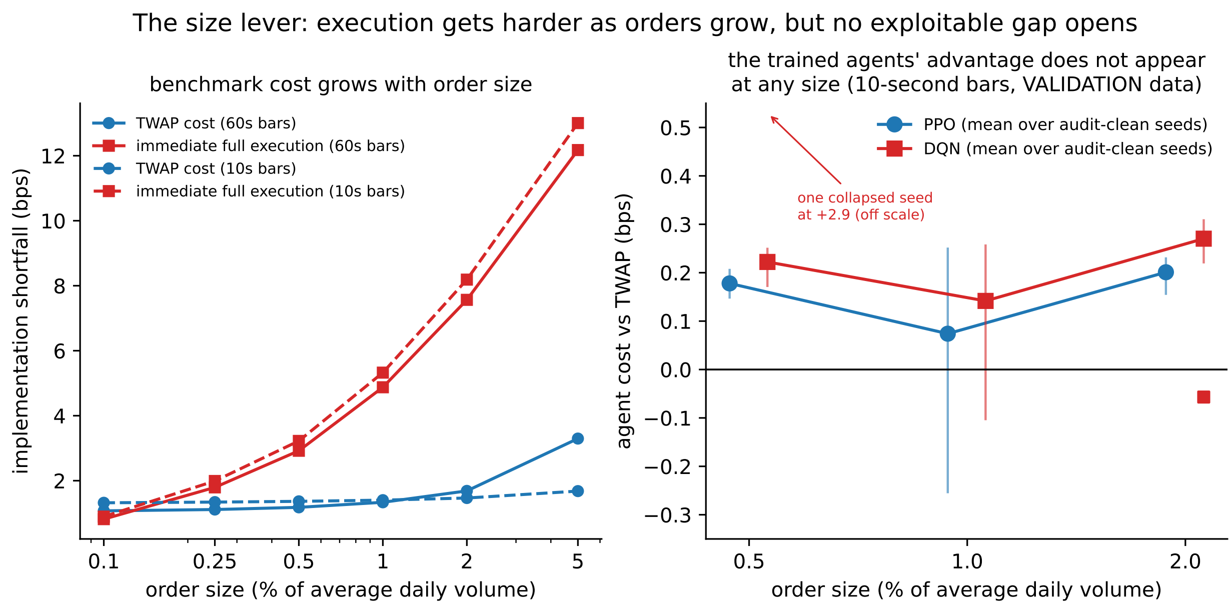

Figure 18: Frozen-replay track size lever (validation data). Left, the cost of two reference schedules versus order size as a share of average daily volume: a TWAP schedule, and immediate full execution, meaning the entire order submitted as one market order at the start. Immediate execution grows steeply with size while TWAP stays comparatively flat, so execution gets harder exactly where scheduling matters. Right, the trained agents' cost versus TWAP at each trained size on 10-second bars; no exploitable gap opens anywhere, with collapsed seeds excluded from the means and annotated separately.

7 Research question 2: regime dependence

Volatility regime shaped every phenomenon in this study, but not in the direction of a usable advantage.

The development-block signal was volatile-only at the primary setting (Figure 3); the grid's spurious triggers were calm-only; and DQN's collapse concentrates in calm markets. The consistent reading is that regime changes the *noise and learning landscape* (calm books recover more slowly relative to volatility, so impact feedback differs) rather than revealing an exploitable edge in either regime: both regimes end at the same confirmed null. The planned attribution study (Section 8) will make the regime mechanism claim precise, and its labelling deliberately waits for the measured-signal extension's verdict.

8 What happens next: the pre-registered programme

The remaining work is committed and ordered, and several of the steps below are already registered in advance as pre-specified experiments. One step from the previous draft, the per-episode cost-distribution analysis for the reactive track, has completed in the meantime and now appears in Section 2. In order:

1. **Frozen-replay track sealed exam (the secondary track).** The test-set evaluator is built next; it must first reproduce known validation numbers exactly (the same bit-exactness standard used elsewhere), and only then is the one-shot exam of all 70 agents run: all 14 arms reported, multiplicity stated, no re-runs.

   Status: The evaluator is under construction and the exam has not yet been run. Its results will convert Table 9 and Figures 17–18 into final test-set results.

2. **Measured-signal extension.** The one remaining experiment that can change the headline: measure the order-flow-imbalance to future-return relationship from the L4 data (magnitude, horizon, decay, per regime), inject exactly the measured relationship into the reactive simulator (drift-neutral, conditioned on the background process, deconvolved from the endogenous channel), and re-run a reduced pre-registered campaign with a fresh sealed block. This unifies the two tracks' mechanisms in one environment.

3. **Liquidity and volume profiles** from the L4 trades (intra-day, intra-week); also supplies the volume curve for a VWAP benchmark.

4. **Almgren-Chriss and VWAP benchmarks**, calibrated to this environment, added to the comparison set.

5. **Attribution (research question 3).** Which order-book features drive the agents' behaviour per regime; scheduled after the extension verdict because its labelling depends on that outcome.

6. **Decision-frequency check.** Environment change (decision interval as a parameter, discount factor re-derived, baselines and audit re-verified, tests), then 12 runs at 0.5-second and 2-second cadences at the primary setting.

A Appendix: full run-level tables

Table 10: Reactive track agent configurations (base and every variant's single change).

code	agent / variant	change from base
base (PPO)	PPO, discrete actions	net 30×5, lr 3e-4, reward ×1, n_steps 2048, ent 0.01, 2M steps
base (DQN)	DQN, discrete actions	net 30×5, lr 1e-4, buffer 1M, ε 1.0→0.01 over 25%
v1a	bigger network	net_arch [64, 64]
v1b	bigger network	net_arch [128, 128]
v2	stronger feedback	reward scale ×100
v3a	faster learning	learning rate 1e-3
v3b	slower learning	learning rate 1e-4
v4a	no exploration bonus	ent_coef 0.0
v4b	more exploration	ent_coef 0.05
v5	longer rollout	n_steps 8192
v6	v2 trained longer	reward ×100, 10M steps
v6b	v3b trained longer	lr 1e-4, 10M steps
w3a	combination	net [128,128] + reward ×100
w3b	combination	lr 1e-4 + reward ×100
d1	DQN collapse fix	exploration_final_eps 0.05, anneal 0.5
d2	DQN collapse fix	reward ×100 + net [64,64]

Table A.1: tuning campaign of the reactive track, all 98 runs per seed (source: `step5_selection_v3/`); the table flows over the following pages.

algo	regime	variant	seed	vs fixed	p	vs adaptive	valid
dqn	calm	_d1	0	+0.0894	0.002	+0.0895	**NO**
dqn	calm	_d1	1	-0.0069	0.780	-0.0068	yes
dqn	calm	_d1	2	-0.0068	0.838	-0.0067	**NO**
dqn	calm	_d2	0	+0.1442	0.002	+0.1443	**NO**
dqn	calm	_d2	1	+0.1393	0.001	+0.1393	**NO**
dqn	calm	_d2	2	+0.0099	0.604	+0.0100	**NO**
dqn	volatile	_d1	0	+0.1398	0.157	+0.1392	**NO**
dqn	volatile	_d1	1	+0.0324	0.307	+0.0318	**NO**
dqn	volatile	_d1	2	-0.0396	0.334	-0.0403	yes
dqn	volatile	_d2	0	+0.0063	0.675	+0.0057	yes
dqn	volatile	_d2	1	-0.0729	0.032	-0.0736	yes
dqn	volatile	_d2	2	-0.0092	0.709	-0.0099	yes
ppo	calm	_v1a	0	-0.0226	0.248	-0.0225	yes
ppo	calm	_v1a	1	-0.0068	0.770	-0.0067	yes
ppo	calm	_v1a	2	-0.0185	0.243	-0.0184	yes
ppo	calm	_v1b	0	+0.0053	0.960	+0.0054	yes
ppo	calm	_v1b	1	-0.0246	0.196	-0.0245	yes
ppo	calm	_v1b	2	-0.0441	0.016	-0.0441	yes
ppo	calm	_v2	0	-0.0111	0.732	-0.0110	yes
ppo	calm	_v2	1	+0.0001	0.787	+0.0002	yes
ppo	calm	_v2	2	+0.0036	0.958	+0.0036	yes
ppo	calm	_v3a	0	+0.0337	0.070	+0.0338	yes
ppo	calm	_v3a	1	-0.0230	0.230	-0.0229	yes
ppo	calm	_v3a	2	-0.0069	0.607	-0.0068	yes
ppo	calm	_v3b	0	-0.0327	0.089	-0.0326	yes
ppo	calm	_v3b	1	-0.0153	0.623	-0.0153	yes
ppo	calm	_v3b	2	-0.0148	0.426	-0.0147	**NO**
ppo	calm	_v4a	0	-0.0189	0.416	-0.0188	yes
ppo	calm	_v4a	1	-0.0126	0.515	-0.0125	yes
ppo	calm	_v4a	2	-0.0298	0.370	-0.0298	yes
ppo	calm	_v4a	3	-0.0382	0.028	-0.0381	yes
ppo	calm	_v4a	4	-0.0076	0.872	-0.0075	yes
ppo	calm	_v4b	0	+0.0027	0.986	+0.0028	yes

algo	regime	variant	seed	vs fixed	p	vs adaptive	valid
ppo	calm	_v4b	1	-0.0161	0.444	-0.0161	yes
ppo	calm	_v4b	2	-0.0043	0.808	-0.0042	yes
ppo	calm	_v5	0	-0.0068	0.808	-0.0067	yes
ppo	calm	_v5	1	-0.0035	0.915	-0.0034	yes
ppo	calm	_v5	2	-0.0377	0.047	-0.0376	yes
ppo	calm	_v6	0	+0.0135	0.671	+0.0136	yes
ppo	calm	_v6	1	+0.0100	0.996	+0.0100	yes
ppo	calm	_v6	2	-0.0099	0.872	-0.0098	yes
ppo	calm	_v6b	0	-0.0319	0.066	-0.0318	yes
ppo	calm	_v6b	1	+0.0047	0.687	+0.0047	yes
ppo	calm	_v6b	2	-0.0053	0.573	-0.0052	yes
ppo	calm	_w3a	0	+0.0017	0.901	+0.0018	yes
ppo	calm	_w3a	1	-0.0178	0.287	-0.0177	yes
ppo	calm	_w3a	2	-0.0029	0.841	-0.0029	yes
ppo	calm	_w3b	0	-0.0218	0.235	-0.0217	yes
ppo	calm	_w3b	1	-0.0102	0.785	-0.0101	yes
ppo	calm	_w3b	2	-0.0032	0.648	-0.0031	yes
ppo	volatile	_v1a	0	-0.0089	0.803	-0.0096	yes
ppo	volatile	_v1a	1	-0.0290	0.374	-0.0296	yes
ppo	volatile	_v1a	2	+0.0232	0.546	+0.0226	**NO**
ppo	volatile	_v1b	0	-0.0150	0.573	-0.0156	yes
ppo	volatile	_v1b	1	-0.0436	0.196	-0.0443	yes
ppo	volatile	_v1b	2	-0.0640	0.071	-0.0646	yes
ppo	volatile	_v1b	3	-0.0870	0.020	-0.0876	yes
ppo	volatile	_v1b	4	-0.0684	0.051	-0.0691	yes
ppo	volatile	_v2	0	-0.0501	0.168	-0.0507	yes
ppo	volatile	_v2	1	-0.0615	0.119	-0.0621	yes
ppo	volatile	_v2	2	-0.0519	0.186	-0.0525	yes
ppo	volatile	_v2	3	-0.0495	0.156	-0.0502	yes
ppo	volatile	_v2	4	-0.0150	0.965	-0.0156	yes
ppo	volatile	_v3a	0	-0.0494	0.243	-0.0500	yes
ppo	volatile	_v3a	1	-0.1044	0.012	-0.1050	yes
ppo	volatile	_v3a	2	-0.0345	0.369	-0.0352	yes
ppo	volatile	_v3a	3	-0.0798	0.023	-0.0805	yes
ppo	volatile	_v3a	4	-0.0427	0.236	-0.0433	yes
ppo	volatile	_v3b	0	-0.0717	0.047	-0.0724	yes
ppo	volatile	_v3b	1	-0.1019	0.006	-0.1026	yes
ppo	volatile	_v3b	2	-0.0759	0.038	-0.0765	yes
ppo	volatile	_v3b	3	-0.0089	0.866	-0.0095	yes
ppo	volatile	_v3b	4	-0.0394	0.344	-0.0401	yes
ppo	volatile	_v4a	0	-0.0501	0.188	-0.0508	yes
ppo	volatile	_v4a	1	-0.0764	0.018	-0.0771	yes
ppo	volatile	_v4a	2	-0.0131	0.944	-0.0137	yes
ppo	volatile	_v4b	0	-0.0402	0.266	-0.0408	yes
ppo	volatile	_v4b	1	-0.0380	0.334	-0.0387	yes
ppo	volatile	_v4b	2	-0.0231	0.422	-0.0238	yes
ppo	volatile	_v4b	3	-0.0492	0.154	-0.0498	yes
ppo	volatile	_v4b	4	+0.0201	0.682	+0.0195	yes
ppo	volatile	_v5	0	-0.0506	0.156	-0.0512	yes
ppo	volatile	_v5	1	-0.0211	0.698	-0.0217	yes
ppo	volatile	_v5	2	-0.0261	0.454	-0.0267	yes
ppo	volatile	_v5	3	-0.0779	0.056	-0.0785	yes
ppo	volatile	_v5	4	-0.0637	0.124	-0.0643	yes
ppo	volatile	_v6	0	-0.0245	0.829	-0.0252	yes
ppo	volatile	_v6	1	-0.0772	0.083	-0.0778	yes
ppo	volatile	_v6	2	-0.0295	0.608	-0.0302	**NO**
ppo	volatile	_v6b	0	-0.0620	0.093	-0.0627	yes
ppo	volatile	_v6b	1	-0.0073	0.976	-0.0079	yes
ppo	volatile	_v6b	2	-0.0474	0.151	-0.0480	yes
ppo	volatile	_w3a	0	-0.0602	0.100	-0.0609	yes
ppo	volatile	_w3a	1	-0.0814	0.029	-0.0821	yes
ppo	volatile	_w3a	2	-0.0423	0.470	-0.0430	yes
ppo	volatile	_w3b	0	-0.0456	0.153	-0.0462	yes
ppo	volatile	_w3b	1	+0.0471	0.467	+0.0465	**NO**
ppo	volatile	_w3b	2	-0.0257	0.647	-0.0263	yes

Table A.2: frozen-replay track, all 70 agents per seed (validation data; DL marks the forced deadline purchase completing more than 10% of episodes); the table flows over the following pages.

panel	arm	seed	vs TWAP (bps)	resid
1-min bars, 30-min	PPO 96.57	0	-0.1707	0.8%
1-min bars, 30-min	PPO 96.57	1	-0.0834	0.2%
1-min bars, 30-min	PPO 96.57	2	-0.1169	0.5%
1-min bars, 30-min	PPO 96.57	3	+0.0487	1.0%
1-min bars, 30-min	PPO 96.57	4	-0.1230	0.5%
1-min bars, 30-min	PPO 193.13	0	-0.0118	2.0%
1-min bars, 30-min	PPO 193.13	1	+0.0094	0.2%
1-min bars, 30-min	PPO 193.13	2	+0.0314	1.8%
1-min bars, 30-min	PPO 193.13	3	-0.0562	1.2%
1-min bars, 30-min	PPO 193.13	4	-0.1214	0.8%
1-min bars, 30-min	DQN 96.57	0	-0.0766	0.0%
1-min bars, 30-min	DQN 96.57	1	+0.0095	0.5%
1-min bars, 30-min	DQN 96.57	2	-0.0517	0.8%
1-min bars, 30-min	DQN 96.57	3	-0.0781	0.0%
1-min bars, 30-min	DQN 96.57	4	+0.6594	37.0% **DL**
1-min bars, 30-min	DQN 193.13	0	-0.0407	1.2%
1-min bars, 30-min	DQN 193.13	1	+0.3014	36.8% **DL**
1-min bars, 30-min	DQN 193.13	2	-0.0553	0.2%
1-min bars, 30-min	DQN 193.13	3	-0.0251	1.5%
1-min bars, 30-min	DQN 193.13	4	+0.0513	12.0% **DL**
10-s bars, 30-min	PPO 96.57	0	+0.2056	0.0%
10-s bars, 30-min	PPO 96.57	1	+0.1587	0.0%
10-s bars, 30-min	PPO 96.57	2	+0.1846	0.2%
10-s bars, 30-min	PPO 96.57	3	+0.1919	0.0%
10-s bars, 30-min	PPO 96.57	4	+0.1488	0.2%
10-s bars, 30-min	PPO 193.13	0	+0.0956	0.2%
10-s bars, 30-min	PPO 193.13	1	+0.1229	0.2%
10-s bars, 30-min	PPO 193.13	2	+0.2496	0.0%
10-s bars, 30-min	PPO 193.13	3	+0.1552	0.5%
10-s bars, 30-min	PPO 193.13	4	-0.2533	5.5%
10-s bars, 30-min	PPO 386.27	0	+0.2294	0.2%
10-s bars, 30-min	PPO 386.27	1	+0.1743	0.0%
10-s bars, 30-min	PPO 386.27	2	+0.1565	0.0%
10-s bars, 30-min	PPO 386.27	3	+0.2174	0.0%
10-s bars, 30-min	PPO 386.27	4	+0.2272	0.0%
10-s bars, 30-min	DQN 96.57	0	+2.9426	97.8% **DL**
10-s bars, 30-min	DQN 96.57	1	+0.1727	7.0%
10-s bars, 30-min	DQN 96.57	2	+0.2329	0.0%
10-s bars, 30-min	DQN 96.57	3	+0.2339	0.5%
10-s bars, 30-min	DQN 96.57	4	+0.2485	0.0%
10-s bars, 30-min	DQN 193.13	0	+0.2441	0.0%
10-s bars, 30-min	DQN 193.13	1	-0.1025	10.0%
10-s bars, 30-min	DQN 193.13	2	+0.0570	0.5%
10-s bars, 30-min	DQN 193.13	3	+0.2547	0.2%
10-s bars, 30-min	DQN 193.13	4	+0.2563	0.5%
10-s bars, 30-min	DQN 386.27	0	-0.0569	27.0% **DL**
10-s bars, 30-min	DQN 386.27	1	+0.3066	2.0%
10-s bars, 30-min	DQN 386.27	2	+0.2209	1.0%
10-s bars, 30-min	DQN 386.27	3	+0.3004	0.0%
10-s bars, 30-min	DQN 386.27	4	+0.2507	0.0%
10-s bars, 10-min	PPO 96.57	0	-0.0262	0.2%
10-s bars, 10-min	PPO 96.57	1	-0.0946	0.2%
10-s bars, 10-min	PPO 96.57	2	+0.1367	0.0%
10-s bars, 10-min	PPO 96.57	3	+0.0450	0.0%
10-s bars, 10-min	PPO 96.57	4	-0.0280	0.5%
10-s bars, 10-min	PPO 193.13	0	-0.1008	1.0%
10-s bars, 10-min	PPO 193.13	1	+0.0741	1.0%
10-s bars, 10-min	PPO 193.13	2	+0.0985	1.5%

panel	arm	seed	vs TWAP (bps)	resid
10-s bars, 10-min	PPO 193.13	3	+0.0373	1.5%
10-s bars, 10-min	PPO 193.13	4	+0.0529	0.5%
10-s bars, 10-min	DQN 96.57	0	+0.2071	26.0% **DL**
10-s bars, 10-min	DQN 96.57	1	-0.0280	0.0%
10-s bars, 10-min	DQN 96.57	2	-0.0231	0.0%
10-s bars, 10-min	DQN 96.57	3	+0.0553	0.2%
10-s bars, 10-min	DQN 96.57	4	-0.0876	2.0%
10-s bars, 10-min	DQN 193.13	0	+0.4414	48.8% **DL**
10-s bars, 10-min	DQN 193.13	1	-0.0236	3.0%
10-s bars, 10-min	DQN 193.13	2	+0.0772	0.0%
10-s bars, 10-min	DQN 193.13	3	-0.0389	0.0%
10-s bars, 10-min	DQN 193.13	4	+0.1193	0.0%

B Appendix: provenance

Every number in this chapter is generated by script from primary result files that are version-controlled with the project repository under `results_archive/` (reactive track in `qrm/`, frozen replay in `l2/`; the paths below are relative to `results_archive/`). The archive holds every scored evaluation, behaviour audit, run configuration, training curve, trained model, and per-episode cost record, so every figure and table in this document can be regenerated from it. Raw exchange data and the derived training datasets are excluded for size; the repository's pipeline scripts rebuild them from Hyperliquid's public data archive.

result	source of truth
Primary campaign (T2, F3, F14)	`qrm/step5_v3/`
Tuning + selection (T3, A.1, F4, F5)	`qrm/step5_selection_v3/`
Sealed confirmations (T4, F6)	`qrm/step5_confirm_{v3a,v1b}/`
Sweeps + grid (F10, F11, T6)	`qrm/step5_sweep_*, qrm/step5_grid_*/`
Replication ladder (F12, T7)	`qrm/step5_esc_*, qrm/step5_xblock_*/`
DQN probe + d3 (F15, T8)	`qrm/step5_dqnprobe_*, qrm/step5_d3_*/`
Q-value diagnostic (F16)	`reports/diagnostics/dqn_q_flatness.json` (repository path)
Environment gates (T1)	`qrm/step4_gates_v3.json, qrm/step3g/`
Frozen-replay agents (T9, A.2, F17)	`l2/runs/, l2/runs_10s/, l2/runs_10s_10min/`
Frozen-replay benchmark sweep (F18)	`l2/size_sweep_results.json`
Learning curves (F7)	`qrm/runs_*/` per-run `curve.json`
Drift confound, before/after (F1)	`qrm/SUPERSEDED_step5_v2/` (pre-fix), `qrm/step5_v3/`
Per-episode distributions (F8, F9, T5)	`qrm/per_episode_v3/`